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We define $f(x, y)$ as number of different corresponding bits in the binary representation of x and y.
For example $f(2,7)=2$ since its binary of 2 an 7 are 010 and 111. The First and third bit differ aso $f(2,7) = 2$.

You are given array of N positive integer A_1, A_n, A_{13}. Find sum of $f(A_i A_j)$ for all pairs (i,j) such that $1 \le i, j \le N$. Return the answer modulo 10^9+7.

Brute force -

S1 - Traverse all the pairs of integers in the array
S2 - Check for both ~~bit~~ integers that ttiere same bit is set or same bit is not set using
$n \& (1<<i)$
S3 - Return answer % (10^9+7)

Or

S1 - Compute $A_i \wedge A_j$
S2 - Count set bits in XOR
S3 - Add to answer.

$O(N^2)$

Optimal -

S1 - For each bit 0 to 31

S2 - Count number having that bit set

S3 - Compute com 0 = N - com 1

S4 - Add 2 × com 1 × cont 0

S5 - Return ans % mod.

```
int solve(Vector<int> &A){
    long long mod = 1000000007
    long long n = A.size();
    long long ans = 0;
    for (int bit = 0; bit < 32; bit++){
        long long count1 = 0;
        for (int i = 0; i < n; i++){
            if (A[i] & (1 << bit))
                count1++;
        }
        long long count0 = n - count1;
        ans = (ans + (2LL * count1 * count0) %
                mod) % mod;
    }
    return ans;
}
```

Complexity

O(N)